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Electroless nickel deposition over the surface of bismuth telluride wafer as a barrier layer

Chunguang Suo, Lei Zhao, Xinghua Shi, and Jinbin Gui

Faculty of Science, Kunming University of Science and Technology, Kunming, China

ABSTRACT

Technology of electroless-deposition nickel on the surface of thermoelectric cooling wafer (Bi_2Te_3) was investigated in the paper. It consists of coarsening, removing coarsened films, reduction, sensitization, activation, pre-reduction, electroless-deposition nickel, heat treating. Bi_2Te_3 is attributed to catalyze poison material to the electroless-deposition nickel bath. We adopted sensitization, activation and reduction to treat wafer, then catalytic activity crystal nucleus which adsorbed the surface of Bi_2Te_3 may rapidly induce nickel-deposited to the whole wafer, so as to keep Bi_2Te_3 form poisoning bath and retarding nickel-deposited. For controlling the thickness of film, the relation between the thickness of film and how long to dispose in the plating bath was observed by tests. At the same time, the process parameters of electroless-deposition nickel bath were discussed. For the complicated working processes are simplified to make it easier to operate and maintain, simplified test processes has been investigated in the paper. Short-cut test processes: before electroless-deposited nickel on the surface of wafer, thin bismuth coating was plating, Nickel coating was obtained by the way.

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Bismuth telluride; electroless nickel deposition; coarsening; activation

1. Introduction

Thermoelectric generator is a well-established technology for compact low power output long-life applications [1]. Thermoelectric materials can produce electrical power in temperature gradients via the Seebeck effect and provide cooling capability via the Peltier effect. Bi_2Te_3-based alloy are the most commercialized thermoelectric materials for applications, typically reaching a thermoelectric figure of merit, ZT, close to unity near room temperature [2]. They have many important applications; for example, used as thermopiles, thermal sensors and thermoelectric coolers for laser diodes [3].

However, thin-film thermoelectric devices with nanostructures still suffer from problems associated with the production such as long process time, high fabrication cost, and small temperature difference across the thin film, etc. On the other hand, the screen printing technique can overcome the production associated issues

CONTACT Chunguang Suo suochunguang@126.com
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because the thermoelectric film can be fabricated in a much cheaper and faster way and the film is thick enough to induce a large temperature difference [4]. When manufacturing thermoelectric coolers, copper is employed as guide baffle because of its superior electrical and thermal conductivity properties. But many experiments show that copper atoms may diffuse into Bi_2Te_3 substrates easily, thus reduce the thermoelectric coolers properties. Electroless nickel protection of copper is common in PCB manufacturing. Electroless nickel-phosphorous has been reported to be significantly more effective than electroplated nickel in preventing copper diffusion.

In order to strengthen bonding force between nickel deposition and wafer, coarsening process was employed. The compound Bi_2Te_3 have a rhombohedral structure at room temperature and are distinguished by a high degree of anisotropy in most of the physical properties.

Instead of mechanical coarsening, the chemical coarsening processes was employed because of the mechanical weakness of the Bi_2Te_3 wafer. After wafers were coarsened, a thin film of dark powder adsorbed over the surface occurred. This dark film is poisonous to the plating solution and cannot be rinsed. For it, removing dark coarsened films methods were designed. The composition and topography of the wafer surface each treatment step were characterized by X-ray photoelectron spectroscopy (XPS) and scanning electron microscopy (SEM) separately.

2. Experimental

2.1. Bi_2Te_3 thermoelectric materials

Bismuth telluride is a narrow band gap semiconductor with energy bandgap of 0.15 eV [5]. Bi_2Te_3 has a rhombohedral unit cell with atoms organizer in Bi or Te planes with hexagonal packing. Bonding between neighboring atoms and planes varies from strong covalent to covalent-ionic to weak van der Waals. As expected from this bonding structure, Bi_2Te_3 is brittle at ambient temperature [6]. Thus it is easy to cleave the crystals along the *c*-plane, the mechanical strength is weak, any treatment to the Bi_2Te_3 wafer should be considered, and its structure can be seen in figure 1. Based on Bi_2Te_3 layered structured the material is brittle.

2.2. Coarsening process

In order to improve the binding force between the nickel deposition film and the bismuth telluride substrates, the surface of the materials needs to be coarsened. The Sulfuric acid was employed as coarsening solution, the formula is shown in table 1. And the reaction of the bismuth and telluride with sulfuric acid is shown in equation 1 and 2.

$$2Bi + 6H_2SO_4 \rightarrow Bi_2(SO_4)_3 + 3SO_2 \uparrow + 6H_2O \quad (1)$$

$$Te + H_2SO_4 \rightarrow TeSO_3 + H_2O \quad (2)$$

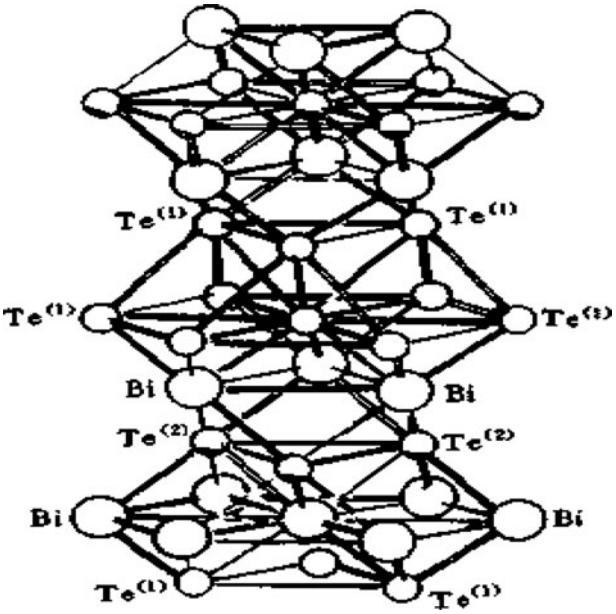


Figure 1. Schematic view of the structure of bismuth telluride.

Table 1. The formula of sulfuric acid coarsening solutions.

content	1	2	3	4
H_2SO_4(98%)/ml	40	50	50	50
H_2O/ml	10	0	0	0
T/°C	Room temperature	Room temperature	Room temperature	Room temperature
t/min	5	5	10	10

2.3. Dark coarsened film removing

After coarsening process a thin dark film appeared to cover the surface of the bismuth telluride substrate. This film will decrease the binding force between the nickel deposition film and the substrates. A multiple acid solution, which is hydrofluoric acid mixed with sodium chromate, is used to remove the dark thin film.

The solution can remove the dark film easily, but the reacting time should be controlled detailed. If the reaction time is too long the dark film can be removed, but at the same time, a new yellow thin film will occur. This thin film can still decrease the binding ability of the nickel deposition over the substrates. In order to remove the dark thin film without forming new other films, four different solutions were used for testing. The formula and the results are listed in table 2.

Table 2. The formula of sulfuric acid coarsening solutions.

Conditions	HF(ml/L)	Sodium chromate(g/L)	Time(second)	Temperature	Results
1	48	125	10	Room temp.	Removal
2	48	125	30	Room temp.	RemovalNew yellow film occured
3	48	250	10	Room temp.	Removal
4	48	250	30	Room temp.	RemovalNew yellow film occured

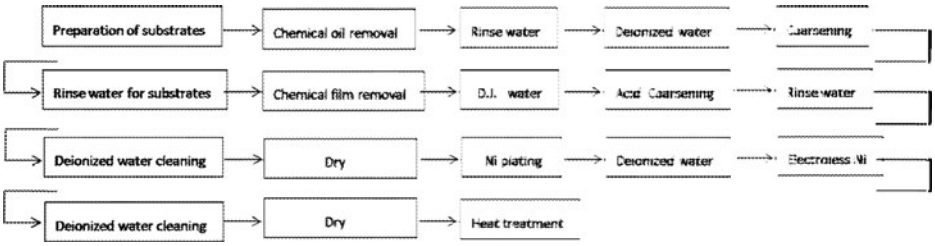


Figure 2. Schematic view of the electroless plating of nickel process.

2.4. Elecctroless plating nickel process

Consequently, electroless-deposited nickel on the surface of catalyze poison material was realized. Sensitization technology: $SnCl_2 \cdot H_2O$ 30∼40 $g \cdot L^{-1}$, HCl(d = 1.18) 50 $ml \cdot L^{-1}$, T room temperature, t 3∼5 min. activation technology: $PdCl_2$ 0.5 $g \cdot L^{-1}$, HCl(d = 1.18) 5∼10 $ml \cdot L^{-1}$, T room temperature, t 5∼10 min. The process of nickel deposition is shown in figure 2.

3. Results and discussions

In order to know morphology of the substrate before and after coarsening process, SEM method was employed, seen in figure 3 and figure 4.

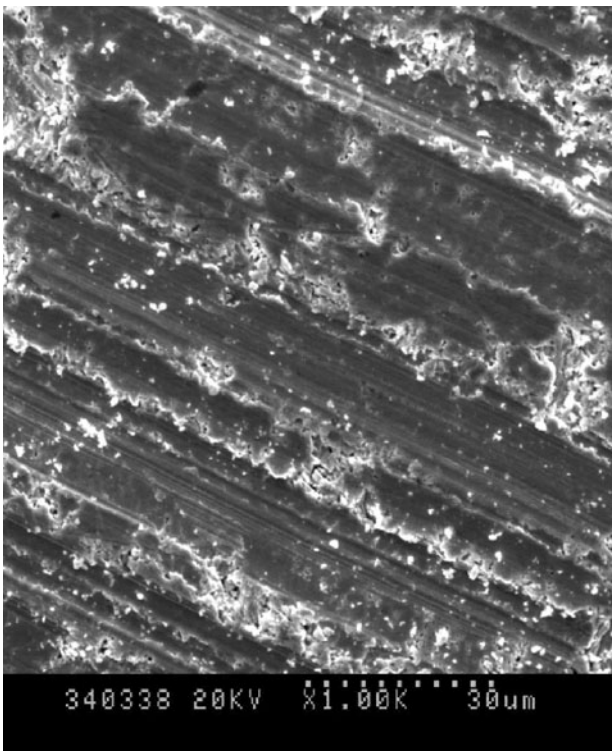


Figure 3. Morphology of surface of the Bi_2Te_3 before coarsening process.



Figure 4. Morphology of surface of the Bi_2Te_3 after coarsening process.

From figure 3 and 4, we can see using sulfuric acid as coarsening sulotion the surface of the substrates were treated well. And the from figure 5–8, the XPS results showed the contents before and after the mixed acid solution treating.

From figure 5–8, the results show that the main contents in dark film are TeO_2, TeO_3, Bi_2O_3, Bi_2Te_3, and thus the solution to remove the dark film can easily formed (table 2).

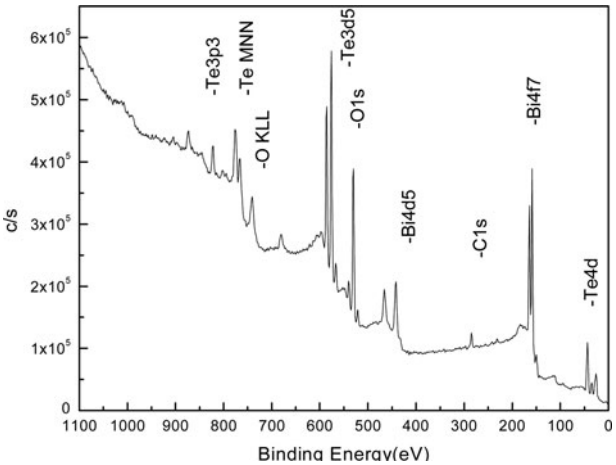


Figure 5. Dark film content.

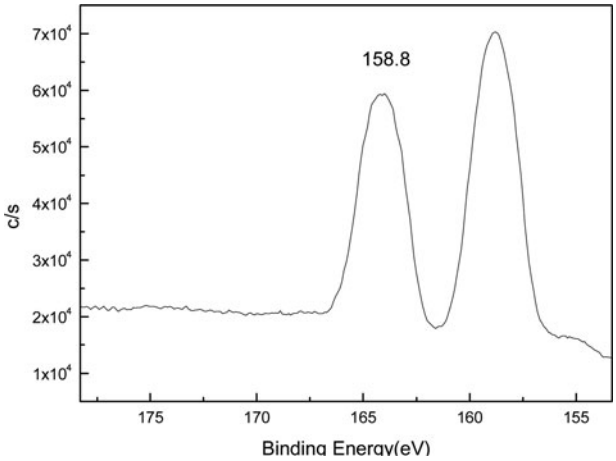


Figure 6. Bi spectrum in dark film.

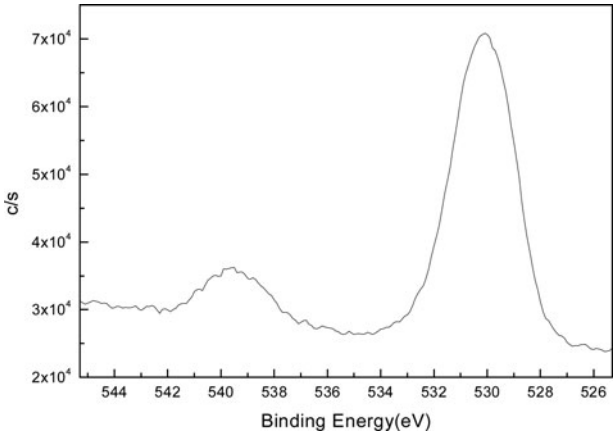


Figure 7. Sb spectrum in dark film.

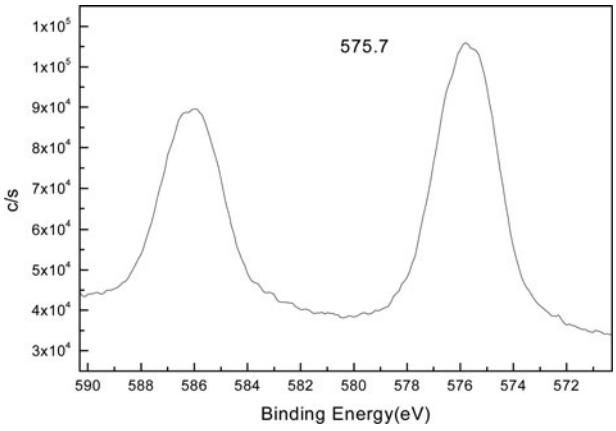


Figure 8. Te spectrum in dark film.

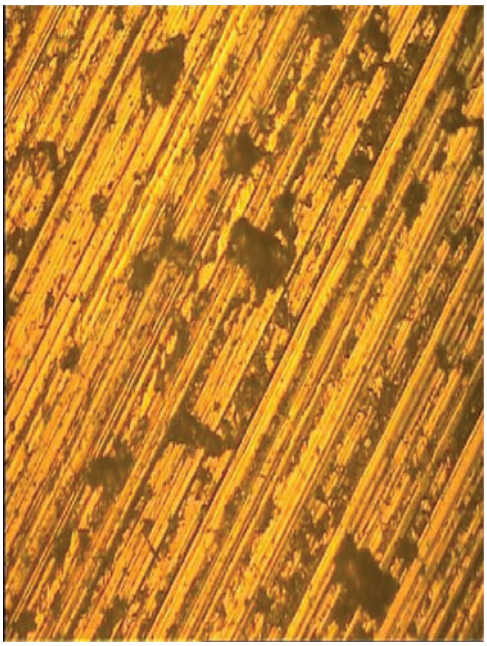


Figure 9. Morphology of Bi_2Te_3 substrate.

Figure 9–12 are morphology of the bismuth telluride substrate before and after treatment of mixed acid solution treat, dark film removal and nickel deposition, separately. Figure 12 showed that after thin dark film removal, the nickel film over the surface is uniform and tightness.



Figure 10. Morphology of Bi_2Te_3 substrate after sulfuric acid oxidation.



Figure 11. Morphology of Bi2Te3 substrate after dark film removal.



Figure 12. Morphology of Bi2Te3 substrate after nickel deposition.

4. Conclusion

In order to strengthen bonding force between nickel plating and wafer, coarsening technology was researched. Lower concentration mixed acid was used. coarsening technology: HNO_3(65∼68%) 100 ml·L^{-1}, H_2SO_4(98%) 120∼150 ml·L^{-1}, T room

temperature, t 1∼3 min. After wafers were coarsened, dark powder film adsorbed on the surface of wafer. The film cannot be rinsed. The composition and structure of corrosion film were analyzed by XRD, SEM and XPS. The results show that coating is composed of TeO_2, Bi_2O_3 and Bi_2Te_3 etc. For it, removing dark coarsened films solution was designed, and then the wafer was reducted in reduction solution. Cleaning surface was acquired. removing coarsened films technology: HNO_3(65∼68%) 100∼150 ml·L^{-1}, CrO_3 20∼30 g·L^{-1}, T room temperature, t 5∼10 min. reduction technology: n type: oxalic acid 30∼50 g·L^{-1}, T 25∼85°C, t 1∼5 min. p type: hydrochloric acid 30∼50%, T room temperature, t 1∼3 min. Heat treating can strengthen bonding force. T 200°C, t 4∼5 h. The thesis has tested bonding force between nickel plating and wafer by heat shock test, thermal cooling cycle examination and welding examination. The results show that bonding force is satisfied with the claim for technology. And pre-reduction was used to enhance product rate and the testing results proved the successful of using the pre-treating method.

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References

1. S. S. Kim, F. Yin, Y. Kagawa, "Thermoelectricity for crystallographic anisotropy controlled Bi-Te based alloys and p-n modules," *J. Alloys and Compounds* **419**, 306–311 (2006).
2. Z. H. Zheng, P. Fan, T. B. Chen, et al, "Optimization in fabricating bismuth telluride thin films by ion beam sputtering deposition," *Thin Solid Films* **520**, 5245–5248 (2012).
3. L. Scidone, S. Diliberto, N. Stein, et al, "Electroless method for Bi_2Te_3 film deposition," *Materials Letters* **59**, 746–748 (2005).
4. Ju Hyung We, Sun Jin Kim, Gyung Soo Kim, et al, "Improvement of thermoelectric properties of screen-printed Bi_2Te_3 thick film by optimization of the annealing process," *J. Alloys and Compounds* **552**, 107–110 (2013).
5. Muralikrishna Molli, Sowmendran Parola, L. A. Avinash, et al, "Solvothermal synthesis and study of nonlinear optical properties of nanocrystalline thallium doped bismuth telluride," *J. Solid State Chemistry* **189**, 85–89 (2012).
6. Zhi-Ping Guan and David C. Dunand, "Compressive creep behavior of cast Bi_2Te_3," *Materials Science & Engineering A* **565**, 321–325 (2013).